Enhance UWB Performance Using Dynamic AI-Enabled Approach

Xu He, *Graduate Student Member, IEEE*, Xiaolin Meng, Lingfei Mo, *Member, IEEE*, Fanxing Yang, Youdong Zhang, and Yue Wu

***Abstract*—Developing more intelligent instrument via artificial intelligence (AI) is highly valuable. Ultrawide band (UWB), with rich radio frequency (RF) and channel impulse response (CIR) features, is widely favored in geospatial applications with AI-based enhancement. This article presents an AI-enhanced UWB solution using a dynamic neural network (DNN) approach, for the improvement of ranging and positioning performance. This dynamic AI-enabled UWB (DAI UWB) solution, employs a pretrained channel estimation model as the policy network to adjust a changeable error fitting network structure for efficient ranging error compensation. Tested in various complex indoor-outdoor environments (up to about 110 m), DAI UWB's performance was evaluated using mixed line-of-sight (LOS)/nonline-of-sight (NLOS) datasets. Comparative results indicate that DAI UWB outperforms the state-of-the-art baselines in ranging and positioning stability. Specifically, in ranging tasks, DAI UWB performs outstanding ranging error compensation while reducing computational waste through dynamic inference routing. Moreover, DAI UWB surpasses all baselines in providing the most stable and accurate positioning results, with the most robust statistical significance. Additionally, DAI UWB achieved a 92.8% accuracy and 93.5% accuracy of channel estimation in real-world and the public benchmark tests, playing a positive role to boost positioning system performance. Furthermore, we believe that DAI UWB's "teaching according to students' aptitude" design philosophy through dynamic AI methods is instructive for developing other AI-enabled advanced instruments.**

***Index Terms*—Channel estimation, dynamic neural networks (DNNs), ranging compensation, ultrawide band (UWB), wireless positioning.**

I. Introduction

NOWADAYS, artificial intelligence (AI) technology is regarded as an enabling technology, which has nonnegligible potential for empowering intelligent Internet of Things (IoT) applications and developing more advanced intelligent instruments [1]. Ultrawide band (UWB) is a wireless measurement technology that is widely applied in various application fields such as positioning, navigation, and spatial perception. Meanwhile, UWB is also an advanced radio frequency (RF) technology. It provides users with a wealth of RF channel characteristics and channel impulse response (CIR) features for channel estimation and ranging error analysis. In recent years, researchers have increasingly recognized the advantages of learning-based methods over traditional approaches. Therefore, they have concentrated on developing representation learning models for channel estimation and ranging error compensation, etc. using RF channel features and CIR features [2].

For example, in wireless positioning systems, nonline-of-sight (NLOS) channel estimation is crucial. Many studies, like [3], [4], have developed learning-based algorithms using the differences in CIR signals and RF channel characteristics between line-of-sight (LOS) and NLOS channels. Some also focus on ranging error compensation by learning the representation between RF channel characteristics or CIR signals and ranging errors through learning-based algorithms like [5], [6]. Moreover, in recent years, there have also been studies that use representation learning methods to compensate for ranging errors while performing channel estimation, such as [7], [8], [9], [10].

Admittedly, these advances have indeed achieved superior performance compared to traditional methods. However, it is important for us to pay attention to the practicality issues of AI-enabled UWB algorithms, namely the contradiction between computational complexity and accuracy. Therefore, developing AI-enabled UWB solutions for channel estimation and ranging error compensation with effective representation and high efficiency in positioning systems with computational timeliness and resource constraints remains challenging. Under this circumstance, we have noticed the capability differences between dynamic neural networks (DNNs) that possess sample-adaptive inference and traditional static deep neural networks that treat everyone alike during inference. This motivates us to find a way out through dynamic AI approaches.

Generally speaking, it is easily understandable that, UWB signals typically exhibit distinct channel response patterns under LOS and NLOS channels. For instance, RF channel characteristics like the first path peak (FP_AMPLI), noise level (STD_NOISE), and the difference between received energy intensity (RX_PWR) and first path energy intensity (FP_PWR) show notable variations between the two channels [11]. Compared with RF channel characteristics, the information density of the CIR sequence is higher, and it is easy to form a different representation between NLOS and LOS channels [4]. Fig. 1 illustrates it.

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The authors are with the School of Instrument Science and Engineering, Southeast University, Nanjing 210096, China (e-mail: xiaolin_meng@seu.edu.cn; lfmo@seu.edu.cn).

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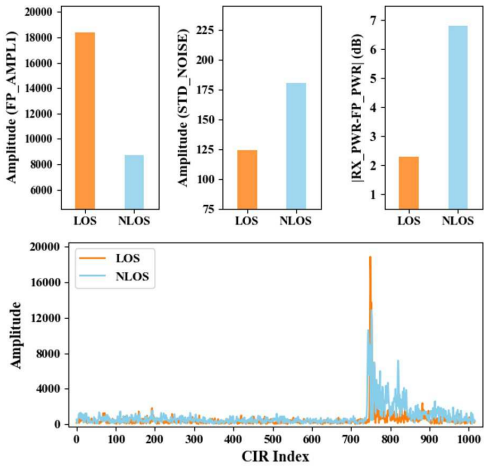


Fig. 1. Pattern differences of UWB RF characteristics under different channels.

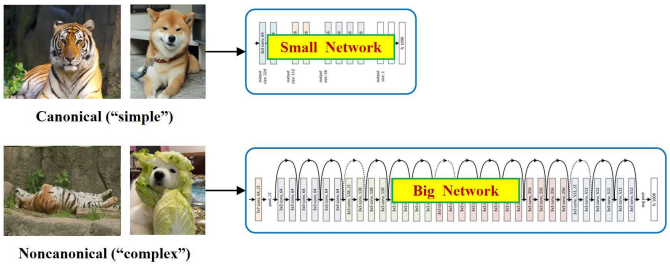


Fig. 2. Sample-adaptive dynamic inference.

This fact implies that although the differences in UWB channel patterns under different propagation conditions make channel estimation relatively straightforward, the UWB signal patterns under NLOS channels are more complex than those under LOS channels, and it is more difficult to conduct effective representation learning of ranging errors. For a representation learning task, the ideal way is to use a smaller-scale model to handle simple patterns at a lower computational cost while ensuring reliable performance. Conversely, a larger-scale model should be employed to process complex patterns to ensure the model's performance. Therefore, a naïve idea is to conduct discriminative representation learning of the data under LOS channels and another under NLOS channels based on the evaluation results of the UWB channel state. That is, using a simple model to compensate UWB ranging errors for LOS channels with simple patterns, while configuring a complex model for complex NLOS channels.

This idea coincides with DNN technology, which uses shallow models for simple samples and more complex models for difficult ones, as shown in Fig. 2. It enables dynamic and efficient sample-adaptive inference [12], a feature lacking in traditional static AI networks [13]. DNN's sample-adaptive inference dynamically adjusts the network structure or parameters based on each input sample. This primarily aims to allocate fewer computational resources to "simple" samples for efficiency or slightly increase the computational load to enhance the model's expressive ability [13].

This article develops an AI-enhanced UWB solution adopting the idea of DNN, called DAI UWB, for the improvement of ranging and positioning performance. Specifically, this article modifies the reported NLOS detection model [3], pretrains and embeds it into the complete DAI UWB model for automatic channel estimation, as a policy network (gate switcher). On this basis, different network structures and weight parameters are, respectively, switched for differentiated representation learning of ranging errors for UWB data evaluated as different channel patterns. Moreover, DAI UWB outputs both the predicted ranging error and channel state. Thus, it is a dynamic AI-enabled solution integrating channel estimation and ranging error compensation. Notably, we believe that the design concept of DAI UWB can also serve as a reference for re-designing other AI-enabled instruments.

To test DAI UWB's capabilities, we gathered extensive data in diverse complex indoor and outdoor environments, creating datasets to train the DAI UWB model and two baselines. These datasets, composed of LOS/NLOS mixed data, include UWB samples with outdoor sampling distances up to 110 m. Notably, most existing UWB studies focus on short-distance, small-scale spatial measurements, with few works addressing UWB measurements over ranges exceeding 100 m [14], [15]. In the ranging task, tested on the two test sets, DAI UWB achieves significant improvement in ranging error compensation through dynamic inference routing. Meanwhile, comparative results demonstrate that DAI UWB's ranging performance matches the state-of-the-art baselines, with higher inference efficiency (see Section IV).

Furthermore, DAI UWB outperforms all baselines in providing the most stable and accurate positioning performance, achieving reductions of 46.4% and 54.2% in comparison with the original positioning errors. Additionally, this article has conducted analyses of statistical significance for the positioning results. Moreover, in NLOS channel detection, DAI UWB demonstrates strong competitiveness when tested on both our self-prepared dataset and the public benchmark, outperforming the most advanced works in recent years (Section IV). The results show that DAI UWB's excellent channel estimation ability gives it an extra edge in positioning tests.

The main contributions are as follows.

1) This article proposes the DAI UWB, a dynamic AI-enhanced UWB solution. The pretrained channel estimation model is designed as the policy network to perform DAI UWB's dynamic routing. This allows the changeable ranging error-fitting network to perform sample-adaptive inference for tailored ranging error compensation according to channel states. This approach may also offer a reference for advancing other AI-enabled instruments.
2) DAI UWB outputs both ranging error predictions and the predicted channel states (LOS/NLOS) of UWB data, forming an AI-enabled solution that integrates channel estimation and ranging error compensation. Its excellent channel estimation also brings extra benefits to strengthen the positioning accuracy. Furthermore, the developed DAI UWB possesses advanced capabilities in

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tasks, like channel estimation, ranging error compensation, and positioning.

3) This article conducted extensive real-world experiments in complex indoor and outdoor environments, assessed DAI UWB's actual performance in NLOS channel detection, ranging error compensation, and positioning tasks, and compared its working performance with that of state-of-the-art baselines through thoughtful design. In addition, this article analyzed how DAI UWB enhances the positioning system's performance through its channel estimation ability.

Outlines: Section II introduces the related work, Section III presents the algorithm design, Section IV shows the performance evaluation and analysis of DAI UWB, Section V is the discussion, and Section VI is the conclusion.

II. Related Work

This article focuses on UWB's channel estimation, ranging error compensation, and positioning, and the core of the proposed DAI UWB lies in DNN technology. This section first reviews the latest UWB advances related to channel estimation and ranging compensation. Then, it briefly revisits the technical points of dynamic AI approaches, focusing on sample-adaptive inference to support the argument that, for a representation learning model, conducting sample-adaptive inference in a targeted manner is rational and advantageous. This also holds true for the development of AI-enhanced UWB technologies.

A. State-of-the-Art of AI-Enhanced UWB Advances

Recent learning-based UWB research has centered on representation learning with RF channel or CIR characteristics to detect NLOS channels and mitigate ranging errors. For instance, Wang et al. [16] and Pei et al. [17] reported strategies using wavelet packet analysis and convolutional neural network (CNN), as well as FCN-Attention for CIR-based NLOS channel detection, respectively. Liu et al. [18] proposed a hard sample meta-learning approach to address CIR-based UWB NLOS channel detection across diverse scenarios and domains. Nkrow et al. [19] utilized transfer learning for CIR-based UWB NLOS detection, generating target domain data via cross-domain mapping and a generative adversarial network (GAN) to create homogeneous features, facilitating knowledge transfer across environments. Moreover, other researchers, such as [20], have also explored the severity of NLOS effects, conducting relevant studies and focusing on detailed classifications of NLOS effects caused by different obstructions. Similar work like [21] analyzed UWB CIR signals and register parameters, employing a XGBoost ensemble learning classifier for effective recognition of NLOS occlusions caused by various media. However, compared to the binary LOS/NLOS classification problem, these multiclassification methods targeting different NLOS effects incur significantly higher data preparation and labeling costs.

Recently, to promote the practicality of AI-enabled UWB technology, we have also developed some lightweight and reliable learning-based methods. For instance, Wu et al. [3] reported a tiny machine learning (TinyML)-based UWB NLOS detection method with effective representation, while He et al. [4] introduced a lightweight learning-based UWB NLOS channel detection method using feature construction techniques. However, all these studies were conducted in relatively short-range settings. Effective UWB channel estimation remains challenging when the distance spans much longer, even exceeding a 100 m. This is due to the common occurrence of obstructions by mixed media and more complex multipath effects in long-distance scenarios.

As is well known, ranging is the foundation of positioning. Works such as that of He et al. [6] and Jiang et al. [22] reported the attention-assisted model and transfer learning-based ranging error compensation solutions. Angarano et al. [23] proposed a robust ranging error compensation method based on edge deep learning and graph optimization for edge devices, which directly uses CIR signals to extract high-level semantics for representation learning. Chen et al. [24] employed an edge deep neural network solution for ranging error compensation and enhanced the UWB positioning performance. Moreover, there are also works that estimate the channel state of UWB while compensating for ranging errors. For example, Yang et al. [7] reported a Fuzzy Transformer-based method, performing representation learning on CIR features and fuzzy probabilities to perform ranging error compensation and channel estimation. Similar works include [8], [9], [10]. Of course, many learning-based studies, like [25], [26], [27], [28], also focused on UWB positioning requirements, proposing to enhance it via NLOS channel exclusion, or direct prediction.

Moreover, the vast majority of the aforementioned state-of-the-art advancements, particularly those that can estimate the channel while compensating for ranging errors, suffer from high computational overhead. However, UWB is, after all, a widely used and lightweight sensor. Therefore, to enhance the practicality of AI-enabled UWB technology, it is highly valuable to develop lightweight but effective learning-based models to mitigate computational resource consumption.

B. Sample-Wise DNNs

So far, powerful deep learning models such as AlexNet, ResNet, DenseNet, and Transformer have emerged. But they use a static paradigm with fixed parameters and structures in both training and inference stages. This leads to the model using the fixed architecture and parameters for all test-stage inferences after training. This limits the representational ability, inference efficiency, and interpretability of the model to a certain extent [12], [13]. Unlike traditional static deep neural networks, DNNs can adaptively adjust their parameters or structure according to sample features or task requirements, enabling adaptive inference. This adaptability endows dynamic AI approaches with significant advantages, such as better performance, computational efficiency, and adaptiveness.

Usually, DNNs can be categorized into three major types: samplewise dynamic networks, spatialwise dynamic networks, and temporalwise dynamic networks. The DAI UWB proposed herein adopts the idea of samplewise dynamic networks. The

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other two types of dynamic networks, which mainly target at images and time-series data, are not discussed here.

The samplewise dynamic networks can dynamically adjust their structures or parameters according to the characteristics of each input sample to achieve data-dependent computation [29]. The primary methods of samplewise dynamic networks include dynamic architecture adjustment (such as dynamic depth and dynamic width) and dynamic parameter adjustment. Dynamic architecture adjustment adapts to the computational demands of different inputs by altering the network's depth (e.g., early exiting [30], layer skipping [31]) or width (e.g., dynamic channel activation [32], [33]). While dynamic parameter adjustment enhances the model's representation power by tuning the network parameters or weights [34], [35], [36], according to the input sample during inference. More details can be found in [12], [13], and [39].

In fact, the core design of sample-adaptive dynamic networks lies in their dynamic routing mechanisms or policy network design. We can borrow the Chinese idiom "teaching students in accordance with their aptitude" to depict their design principle. For instance, cases like [37], [38], [39], [40] showcase that the dynamic AI models with well-designed dynamic strategies can surpass the static AI models in representational ability and inference efficiency, etc.

Currently, DNNs have been applied in fields such as computer vision (CV), natural language processing (NLP), fault diagnosis, and intelligent control, as summarized in [39]. Moreover, dynamic AI models can dynamically adjust the use of computational resources, thereby enhancing the response speed and energy efficiency of edge devices. This advantage is particularly suitable for the widespread application of UWB technology in intelligent mobile applications, forming a dynamic AI-enhanced solution.

III. Algorithm Design

As previously noted, sample-adaptive dynamic inference is more rational for representation learning models than the one-size-fits-all strategy of static AI models. The DAI UWB employs the design concept of sample-adaptive dynamic networks, centered on a policy network that implements dynamic routing, adjusting the parameters and structure of the changeable ranging error fitting network to enable sample-adaptive inference.

Considering the characteristics presented by the RF and CIR signals of UWB, the most intuitive idea is to utilize the channel estimation network to design DAI UWB's policy network, because the signal patterns under NLOS channels are usually more complex. Hence, our previously reported UWB NLOS detection model [3] is modified to construct DAI UWB's policy network in this study. This model uses both RF channel and CIR characteristics to learn the effective channel state representation but was only evaluated on an indoor dataset (eWINE). In this article, it is redesigned into a dual-branch structure with separate RF and CIR inputs, similar to [4], for channel estimation. Then, via pretraining, the policy network is embedded into the complete DAI UWB model to automatically assess each input signal and determine whether its channel attribute is "simple" or "complex."

TABLE I
Comparison of Evaluation on the eWINE Dataset

	Strategy	Accuracy	Precision	Recall	F1 score
1	CNN [25]	87.4%	85.9%	89.4%	87.6%
2	CNN-LSTM [27]	82.4%	79.4%	87.4%	83.2%
	FCN [17]	86.3%	83.8%	89.9%	86.7%
3	LSTM-FCN [17]	86.4%	83.9%	90.0%	86.8%
	FCN-Attention [17]	88.2%	85.9%	91.6%	88.6%
4	***Ours [3]***	***93.1%***	***93.2%***	***93.1%***	***93.2%***
5	***Ours (This Work)***	***93.5%***	***94.5%***	***92.2%***	***93.3%***

Given the effective compensation of ranging error using only RF channel characteristics in our prior work [6], and the low-dimensionality of input features that helps cut model computation costs, for samples assessed as "simple" (i.e., LOS channels) by DAI UWB's policy network, only the RF channel input branch is used. It combines with a simpler ranging error fitting network via cross-layer connections controlled by *Gate1* to compensate for ranging errors in "simple" samples. Conversely, for "complex" samples, as assessed by the policy network, the model uses features from both RF and CIR input branches and employs a deeper, more parameter-rich fitting network, controlled by *Gate2*, for representation learning. Fig. 3 illustrates the DAI UWB model, with details of each functional module provided below.

A. Dual-Branch Input Block

As shown in Fig. 3, the DAI UWB model has two input branches. One is an 11-length vector composed of all UWB RF channel characteristics. The other is the 50-key CIR sequence signals after feature selection. As per [41], the CIR sequences in the special range are crucial for NLOS channel recognition. Thus, this article adopts the same feature selection strategy for CIR signals as [3]. Namely, the 50-length key CIR sequence group is intercepted from FP_IDX − 2 to FP_IDX + 47 in the RF channel characteristics, consistent with [3]. In addition, descriptions of all RF channel characteristics (including the FP_IDX) involved in the 11-length vector of *Input Channel1* can be found in [6, Table I] and won't be elaborated here.

As for the calculation and description of RX_PWR and FP_PWR among the above RF channel characteristics, you can refer to [22, eqs. (3) and (4)]. Moreover, in the self-attention network structure, using dual-branch input channels for different semantic information can also reduce the model's computational load to a certain extent [4].

B. Policy Network for Automatic Channel Estimation

For *Input Channel1*, the Policy Network Block embeds a Soft Attention Block to learn the contribution of different RF channel characteristics to the effective representation of the UWB channel state. The fully connected (FC) layer for linear mapping has 11 neurons with a linear activation function. The FC layer in the soft attention module also has 11

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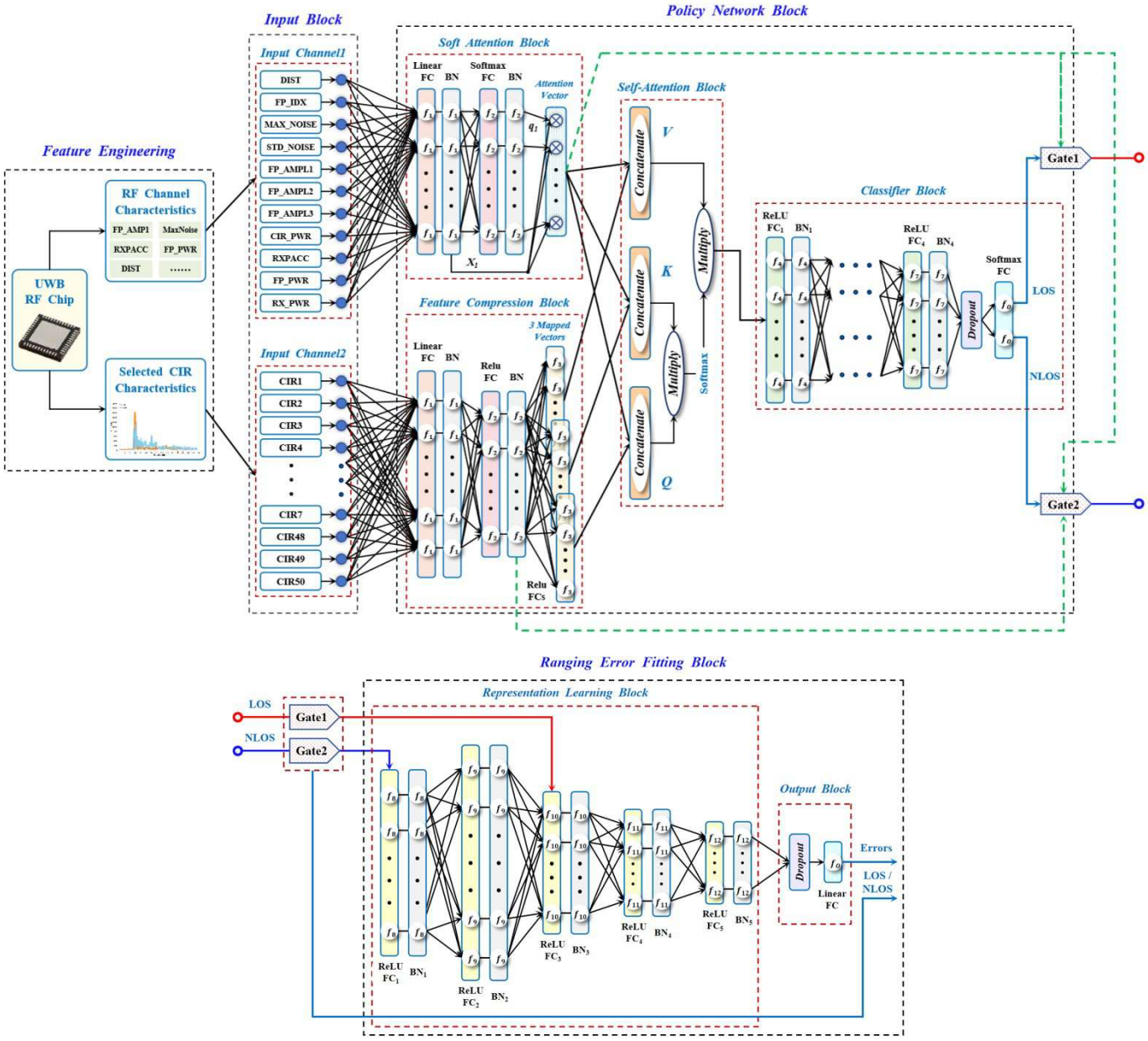


Fig. 3. DAI UWB model.

neurons but uses softmax as the activation function. The batch normalization (BN) layer is utilized to maintain stable data distributions across each layer and accelerate convergence, thereby facilitating easier model learning. The soft attention mechanism is implemented using the dot product attention model, according to [6, eqs. (1)–(3)].

For Input Channel 2, the policy network block employs a multilayer perceptron (MLP) structure as the feature compression block for CIR feature embedding, extraction, and compression. It first uses a linear-activated FC layer with a BN layer for linear mapping to access the 50-length key CIR features. Then, a ReLU-activated FC layer with 30 neurons is connected with the BN layer to further extract features and reduce feature dimensionality. Next, the reduced-dimension features are mapped to three different FC layers, aiming to achieve differential representations of the extracted abstract features. The FC layers for different mapping spaces all use the ReLU activation function and have 14 neurons.

Considering the dual-input branch design, the soft attention vector and the three mapped vectors, housing in different branches, are separate. To fully utilize these two types of semantic features for automatic channel estimation, the policy network first concatenates the attention vector and three mapping vectors from different branches to form the Query (Q), Key (K), and Value (V) feature vectors, as shown in Fig. 3. Then, following [3], a self-attention block is embedded to enhance effective representation learning. Mathematical description of self-attention mechanism can be found in [3, eqs. (1) and (2)].

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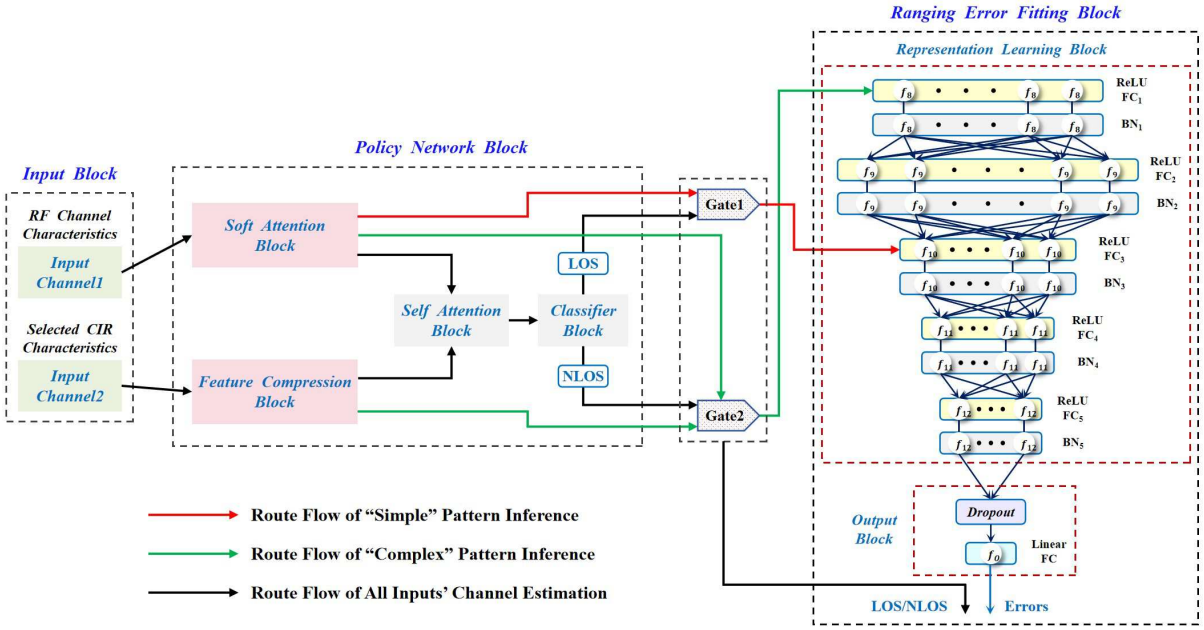


Fig. 4. Dynamic routing of DAI UWB during inference.

On this basis, a classifier block follows the self-attention block, composed of four sequential ReLU-activated FC layers with four BN layers, a dropout layer, and a binary output FC layer with Softmax activation. These four FC layers have 64, 32, 16, and 8 neurons, respectively, and the Dropout ratio is 0.25. To enable the abovedescribed network structure to function as a policy network in the complete DAI UWB model, its evaluated binary state output is used as the dynamic routing gates activation basis.

This article adopts a pretraining strategy for the policy network. That is, we first train the network comprising the input block and policy network block. Afterward, we freeze the weights of the input block and policy network block to embed them into the complete DAI UWB model for automatic channel estimation. The following is a description of the pretraining process.

During the policy network training, this article sets the epoch to 80, the batch size to 30, uses "sparse categorical cross-entropy" as the loss function, and Nadam [42] as the optimizer. The description of the dataset used for model training can be found in Section IV. In the stage of model training, LOS samples are labeled "0" (simple mode), while NLOS samples are labeled "1" (complex mode). Posttraining, weights of the entire policy network block and input block are frozen, excluding them from further training. After embedding the policy network into the DAI UWB model, the attention vector in the soft attention block, the BN layer's output tensor preceding the three mapped vectors in the feature compression block, and the output tensors of the classifier block are set as accessible returns. This configuration is necessary for the design of dynamic routing for the changeable ranging error fitting block, enabling the DAI UWB model to output channel estimation results and perform sample-adaptive inference.

C. Ranging Error Fitting Network With Dynamic Gating

As shown in Fig. 3, the policy network's output dictates the activation states of *Gate1* and *Gate2*. Based on the policy network's estimation results, discerned via the index value of the binary-output tensor, "simple" and "complex" samples are routed to the different network structures controlled by *Gate1* and *Gate2* for ranging error fitting, respectively. Since the gating logic is very simple to implement in the TensorFlow and PyTorch frameworks, this article only explains the network structures for performing the ranging error fitting tasks.

The ranging error fitting block shown in Fig. 3 is a typical deep neural network structure comprising two subfunctional blocks. During DAI UWB inference, its activation is controlled by *Gate1* and *Gate2*. The representation learning block consists of FC layers with 64, 128, 32, 16, and 4 neurons in sequence, each followed by a BN layer, and all with ReLU activation. The *Output Block* includes a 0.1-ratio Dropout layer and a linear FC layer for predicted ranging error output.

To facilitate readers' understanding of how DAI UWB implements sample-adaptive inference in a manner of "teaching students in accordance with their aptitude," Fig. 4 illustrates the dynamic routing of DAI UWB during the inference. Unlike [8] and [10], which use parallel neural network structures, DAI UWB employs an integrated structure to perform channel estimation and ranging compensation. For each input, DAI UWB first conducts LOS/NLOS classification along the black route flow depicted in Fig. 4. Samples classified as LOS are treated as "simple" patterns, while those classified as NLOS are treated as "complex" patterns, thereby determining the activation states of *Gate1* and *Gate2*. Additionally, regardless of which gate is activated, the channel estimation results for LOS/NLOS classification are output by the DAI UWB model. As noted earlier, the output tensors from the soft attention block and feature compression block

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in the policy network block are accessible, enabling them to combine with different gates and parts of the variable ranging error fitting block to form a dynamic network graph.

For "simple" samples, the policy network's output layer returns index "LOS," activating *Gate1*. This triggers the representation learning block's last three groups of FC and BN layers, plus the whole output block, within the ranging error fitting block, with corresponding weights loaded. The 32-neuron FC layer connects cross-layer with the policy network's attention vector from the soft attention block. Here, the DAI UWB model merely uses the 11 RF channel features to form the input vector (*Input Channel1*) for training and inference. During inference, the total parameter count is 12 217, including 1257 for the ranging error fitting block.

For "complex" samples, *Gate2* is triggered to activate all layers in the complete ranging error fitting block, with corresponding weights loaded. The attention vector from the soft attention block and the output tensor from the feature compression block in the policy network are concatenated as input for the ranging error fitting block. In this case, the DAI UWB model uses all RF channel and CIR features for inference, with a total parameter count of 29 985 during inference.

Training details are as follows. As previously detailed, the dataset is precisely labeled, with ground truth (GT) sourced from a real-time kinematic (RTK) device or laser rangefinder. During training, the weights of the input block and the policy network block are frozen to autonomously determine the gating activation status and execute corresponding weight updates. Training uses the mean squared error (mse) loss function, Nadam optimizer, and the epoch and batch size are 50 and 30, respectively.

IV. Evaluation and Analysis

Many current studies view UWB as a short-range radio technology for indoor or limited-area spatial perception and measurement. Yet, UWB technology has advanced, with commercial modules supporting ranging over hundreds of meters. Few studies explore UWB measurement from tens to hundreds of meters, especially in diverse complex indoor and outdoor environments. To evaluate DAI UWB's real-world performance, this article tested it in terms of channel estimation, ranging error compensation, and wireless positioning tasks in various challenging indoor and outdoor settings.

Additionally, since [25] provides the eWINE dataset, an open-source dataset dedicated to UWB channel estimation, this article also documents the evaluation and comparison of DAI UWB on it. It is worth mentioning that the CIR data format provided by DW1000-developed UWB devices is related to its pulse repetition frequency (PRF) setting. When the PRF is set to 64 and 16 MHz, the CIR signals are provided as sequence of 1016 real numbers or 992 complex numbers, respectively [11], [43]. Although many works have used UWB devices with a PRF setting of 64 MHz to collect CIR signals for research and analysis, it is currently very difficult to find an open-source dataset that matches the CIR features of this article (PRF = 64 MHz) and can be used for ranging error compensation. Therefore, the ranging error compensation experiments in this section were all tested on our self-prepared dataset. It should be stated here that the method proposed in this article is applicable to UWB devices developed based on the DW1000 chip and its latest subsequent products (such as DW3000), provided that the CIR data format conforms to the setting of PRF = 64 MHz.

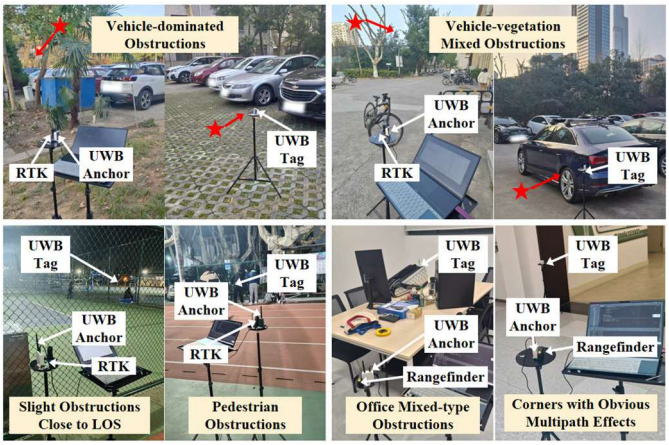


Fig. 5. Some representative dataset collection records. The red pentagram markers indicate the obstructed UWB tags within the same set of measurements.

Note. Data can be obtained from the main authors for legitimate reasons.

A. Data Preparation

The self-prepared dataset was collected in diverse complex indoor and outdoor environments. Each sample has channel state and ranging error value annotations. The dataset contains 8808 NLOS and 6362 LOS samples, with the farthest sampling distance exceeding 110 m. The training set has 7617 data samples (4399 NLOS and 3218 LOS samples), while the validation set has 3720 samples (2179 NLOS and 1541 LOS samples) and the test set has 3833 samples (2230 NLOS and 1603 LOS samples). The training and validation sets have no data overlap, and the NLOS samples come from environments with obstacles such as vehicles, vegetation, people, and roadside furniture, as shown in Fig. 5.

The positioning-task dataset in this section's results is independent of the training and validation sets and can serve as the second test set, containing 1395 samples. Its sampling environments also include diverse complex indoor and outdoor settings. Indoor environments include office furniture, electronic devices, daily necessities, personnel obstructions, spacious halls (about 100 m^2), and narrow corridors prone to multipath effects. Outdoor settings encompass open sports fields, sparsely populated areas, and scenes with vehicles, people, vegetation, and mixed-type obstructions.

Notably, the UWB modules are proximity to the ground due to limited tripod height at measurement distances of tens to over 100 m, resulting in some samples being affected by ground-reflection multipath effects. These multipath-contaminated LOS channels challenge DAI UWB's accurate channel estimation, as their CIR signals may exhibit NLOS characteristics without significant impact on RF channel

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features. Moreover, this type of data has a negative impact on ranging error compensation and positioning tasks, due to its small proportion and remarkably skewed distribution. However, such data are inevitable in wireless technology applications. Thus, this article does not exclude them, and they exist in small proportions in the training set, validation set, and two test sets.

In addition, the second test set containing mixed LOS/NLOS samples was gathered during unplanned, random movement while conducting the positioning task, so only coordinate information was recorded without channel-state annotations. The above is the description of the self-prepared dataset for testing the DAI UWB model and baselines.

B. Baselines Introduction

Next, the baselines employed in this article are presented. To our knowledge, it seems that no related work exists in dynamic AI-enabled UWB. Hence, this article utilized some advanced static end-to-end models for comparative evaluation.

1) Baseline1: This article modifies the prior attention-assisted ranging error compensation algorithm [6] as Baseline1. This method has delivered advanced ranging error compensation in measurements spanning several to over ten meters, effectively suppressing multipath reflections. Moreover, assessed under identical conditions, it outperforms the transfer learning-based approach reported in [22]. Its detailed description is as follows.

Baseline1 comprises *Input Channel1* of the DAI UWB model, the soft attention block presented in the policy network, and a deep network for ranging error fitting. Given that Baseline1 uses merely RF channel characteristics for end-to-end ranging error representation learning, this article configures a deep neural network with more layers for UWB ranging error representation fitting, embedded after the soft attention block. The specific structure consists of six sequential FC layers with ReLU activation, containing 32, 64, 96, 32, 16, and 4 neurons, respectively. Each FC layer is followed by a BN layer to accelerate training. Subsequently, a Dropout layer with a 0.1 dropout rate and a linear-activated FC layer with single neuron for prediction output are appended. During training, the model employs the mse loss function, the Nadam optimizer, and is trained for 50 epochs with a batch size of 30.

2) Baseline2: The second baseline adopted herein is the policy network modified to perform the ranging error fitting task. Namely, the original Softmax-activated output layer for binary classification is replaced with a linear-activated FC layer with a single neuron to perform the ranging error fitting task. Besides the output layer, the remaining structure of Baseline2 consists of six sequential ReLU-activated FC layers with BN layers, containing 32, 128, 64, 32, 16, and 8 neurons, respectively. The relevant training configurations are exactly the same as those of Baseline1 to facilitate performance evaluation and comparison under the same conditions.

3) Baseline3: The third baseline utilizes a CNN-LSTM-based neural network, a welcomed architecture that combines the strengths of CNN and long short-term memory (LSTM) and is widely applied in CV, NLP, and various engineering applications, including UWB-related studies like [27] and [28]. The structure of Baseline3 is composed of three main parts. The first part is an input layer that accepts a 61-length 1-D vector, comprising all RF channel and CIR features. The second part is composed of two sets of Conv1D and MaxPooling layers followed by two LSTM layers. Where the filter sizes of two Conv1D layers are 64 and 128, respectively, with a kernel size of 3 and ReLU activation, while the two LSTM layers have unit sizes of 64 and 32, respectively. The third part is an MLP structure, including two sets of ReLU-activated FC layers (with 64 and 32 neurons, respectively) with BN layers, followed by a Dropout layer (with a ratio of 0.1), and finally an output FC layer with linear activation and a single neuron.

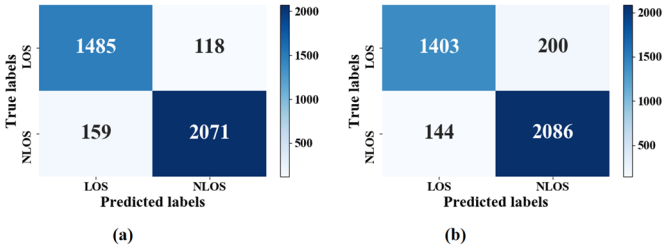


Fig. 6. Confusion matrixes on the first self-prepared test set with channel state annotations. (a) and (b) DAI UWB and Reference [3], respectively.

C. Visualization and Evaluation

First, this article evaluates DAI UWB's performance in the NLOS channel detection task. Fig. 6 presents the confusion matrix of DAI UWB tested on the first test set. Here, correctly predicted NLOS samples are labeled as true positives (TPs), while incorrectly predicted NLOS samples are labeled as false positives (FP). Correctly predicted LOS samples are labeled as true negatives (TNs), and the remaining incorrect predictions are labeled as false negatives (FNs).

Tested on the first self-prepared test, the results indicate that DAI UWB achieves an accuracy of 92.8%, precision of 94.6%, recall of 92.9%, and $F1$ score of 93.7%. Under identical testing conditions, our recently reported work [3], which outperforms numerous leading baselines in fair evaluations, achieves an accuracy of 91.0%, precision of 91.3%, recall of 93.5%, and $F1$ score of 92.4% on the first test set.

The eWINE dataset was sampled across seven distinct indoor scenarios, with each scenario retaining 3000 NLOS samples and 3000 LOS samples [25]. In this article, the eWINE dataset was divided into training, validation, and test sets in a ratio of 5:1:1. Table I records the related metrics on the eWINE dataset.

It is not difficult to see from the intuitive comparison results in Table I that the strategies we reported have a clear advantage in the channel estimation task. DAI UWB's performance is relatively close to that of [3] in various indicators, with a small improvement in all indicators except recall. Specifically, it achieved an accuracy of 93.5%, a precision of 94.5%, a recall of 92.2%, and an $F1$ score of 93.3%. The pretrained policy network of DAI UWB for channel estimation is restructured from [3] into a dual-branch input architecture in this article

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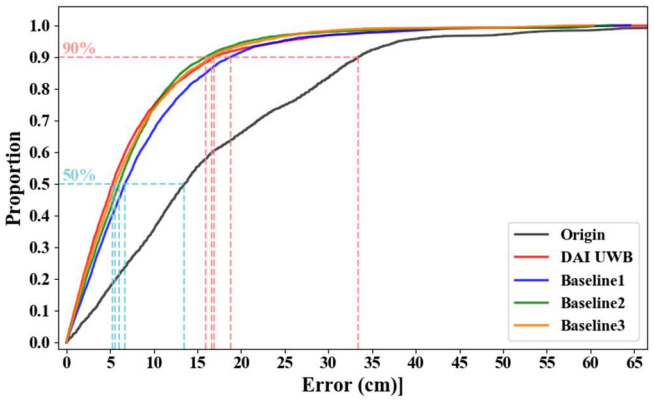


Fig. 7. Cdf plots of ranging error compensation for each strategy on the hybrid test set.

TABLE II
COMPARISON OF RANGING ERROR COMPENSATION EFFECTS ON THE HYBRID TEST SET

Unit: cm	Origin	DAI UWB	Baseline1	Baseline2	Baseline3
90% line	33.5	16.9	18.8	16.0	16.6
85% line	30.7	14.2	16.0	13.2	13.8
75% line	25.1	10.1	12.2	10.2	10.4
50% line	13.5	5.3	6.7	6.0	5.6
25% line	7.0	2.4	3.2	2.8	2.6

to fit the dynamic routing mechanism for sample-adaptive inference shown in Fig. 4. In fact, it may refer to [3] to see that, compared with other advanced works reported in recent years, our strategies have a significant lightweight advantage in terms of model parameter scale, besides the advantage in accuracy.

This article merges the two self-prepared test sets to form a hybrid test set for evaluating the ranging performance of DAI UWB. Fig. 7 shows the ranging error compensation performance of DAI UWB and various baselines on the hybrid test set using plots of the cumulative distribution function (cdf) for intuitive comparison. It shows that DAI UWB reduces the 90% error line from 33.5 to 16.9 cm, the 75% error line from 25.1 to 10.1 cm, and the median error from 13.5 to 5.3 cm. Apart from Baseline1, the performance of the other strategies is quite close.

Table II lists each strategy's error compensation results on the hybrid test set. Although Baselines3's parameters and computational cost are significantly higher than those of other strategies, its performance on the hybrid test set is not the most outstanding. It thus appears that merely increasing the computational complexity of the model does not necessarily achieve the goal of significantly enhancing the representation capability of UWB ranging error.

Furthermore, analyzed using quantile-quantile (QQ) plots, we found that the original ranging errors, compensated by DAI UWB and the three baselines, all exhibit normal distribution characteristics. Therefore, to further enhance the statistical features of the data analysis, Fig. 8 illustrates the statistical characteristics of ranging error distributions under different strategies. Fig. 9 ignores the histogram background of the ranging errors and focuses on displaying the normal statistical characteristic curves corresponding to the different strategies.

The visualizations in Figs. 8 and 9 indicate that, compared with other baselines, the distribution of ranging errors after compensation by DAI UWB is the most compact, as evidenced by the corresponding normal distribution having a mean value closest to zero and the smallest variance with the smallest skewness. Moreover, this article also conducted a T-test on the normality characteristics of all strategies. Since their p-values are all close to 1.0 and significantly greater than the empirical value of 0.05, it can be concluded that the ranging error distributions caused by these strategies conform to the normal distribution characteristics labeled in Fig. 8.

Finally, we present the testing and evaluation of DAI UWB in the positioning task. The data description is detailed in the data preparation. The positioning experiments took place in 13 diverse and complex outdoor settings ranging from around 4–105 m, and seven indoor settings with distances spanning 2.5–6.2 m. In each scene, UWB tags were fixed and linked to 6–8 groups of sampled UWB anchor positions and UWB measurements. The positions of the UWB anchors during the localization experiments were completely random. During the localization data collection process, UWB anchors were kept stationary at different spatial locations, and a small number of data samples were collected accordingly. These samples constitute the second test set described earlier.

This article employs the principle of range intersection [44] to construct the localization model, incorporating ranging error compensation, as shown in the following equation:

$$f_{f^{obj}} = \sum_{i=1}^{n} \sqrt{(x - x_i)^2 + (y - y_i)^2 - (d_i + \delta_i)} \tag{1}$$

where n is the total number of anchor positions. (x, y) and (x_i, y_i) denote the unknown tag coordinates and the ith anchor coordinate, respectively. d_i and δ_i represent the raw ranging observation and its compensated error component. The L-BFGS-B algorithm [45], a quasi-Newton method, was used to solve the nonlinear least squares (NLSs) positioning system presented in (1).

As previously mentioned, multipath effects in UWB measurements are unavoidable and negatively impact the positioning system. Upon reviewing the channel estimation results for each set of UWB data sampled under static working mode, an interesting phenomenon was observed. This led us to the decision to output predicted ranging errors, as well as channel estimation results, when designing the DAI UWB model. The phenomenon is described as follows.

Taking a group of data sampled for UWB tag localization as an example, the data were collected under LOS propagation conditions between the tag and anchor, where the tag was placed at a corner. Due to multipath reflections from walls and corners, some LOS samples exhibit a "complex" CIR pattern, as shown in Fig. 10. When DAI UWB performs inference on these data, most samples are classified as "simple" (LOS channels), while the contaminated ones are deemed "complex." This observation indicates that DAI UWB can filter out some channels significantly contaminated by multipath effects in static UWB positioning systems.

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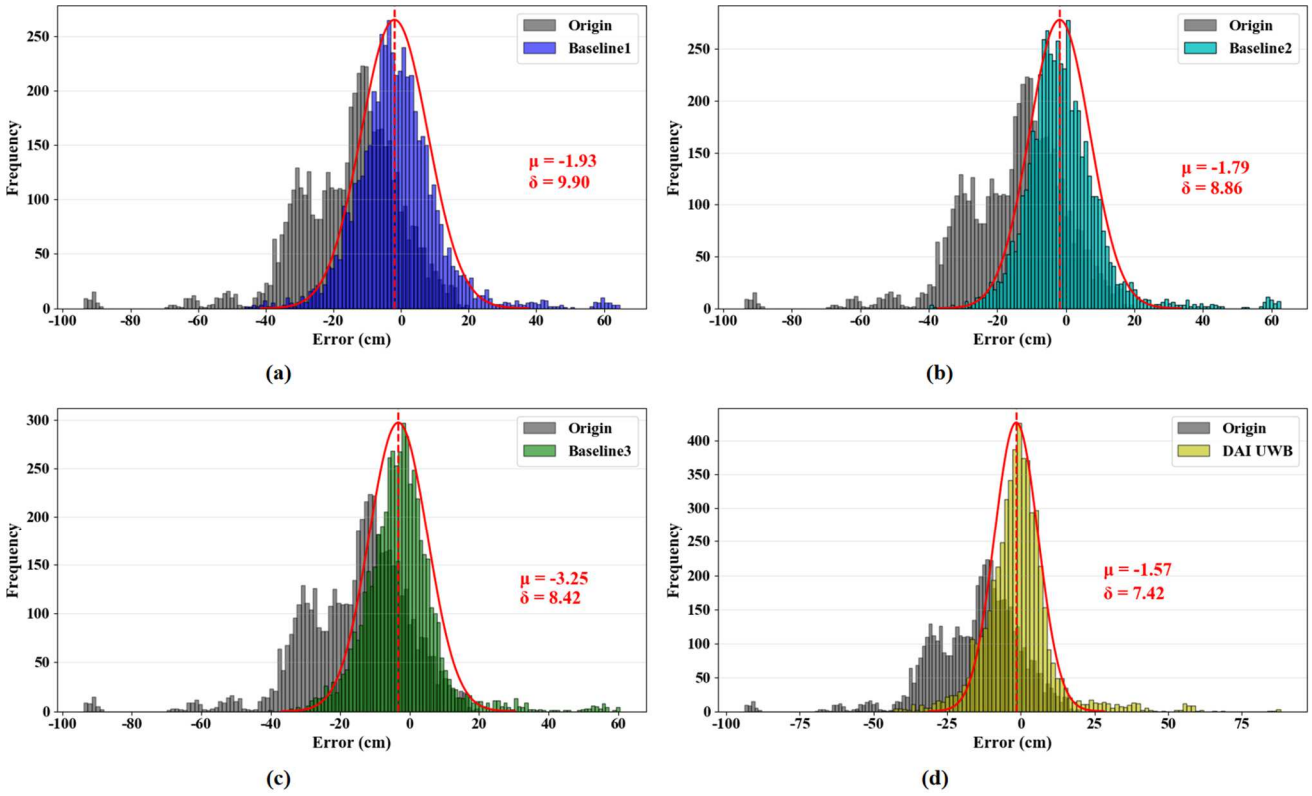


Fig. 8. Histograms of error distributions for various strategies fit with normal distribution curves. Where, μ and δ are the mean and standard deviation of the normal distribution (a)–(d) Baseline1, Baseline2, Baseline3, and DAI UWB, respectively.

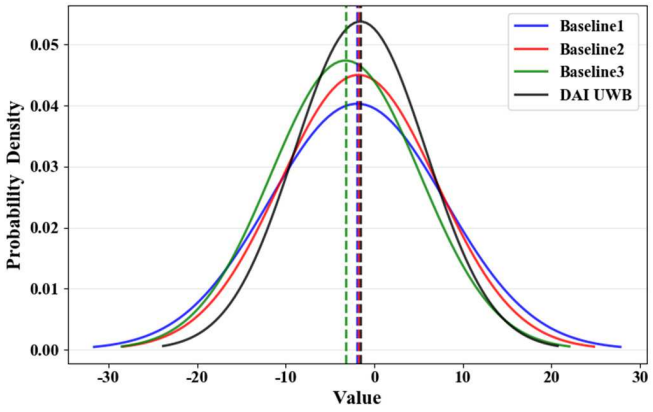


Fig. 9. Illustrations of normal statistical characteristic curves.

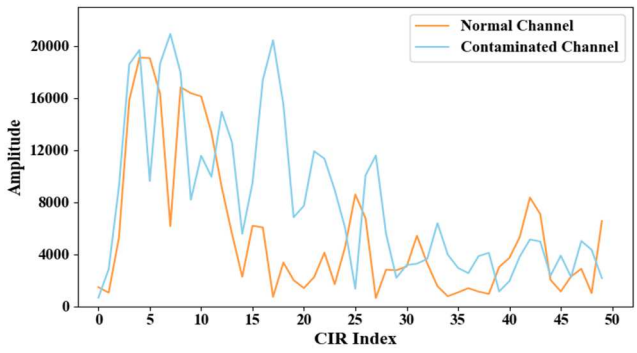


Fig. 10. Example of phenomenon explanation.

To visually compare the positioning performance of DAI UWB-enhanced channel cleaning with the original, uncleaned, contaminated channel, this article plots the shooting target charts depicted in Fig. 11. The center of the target represents the GT of the tags, while different colored target areas indicate the error bounds corresponding to varying positioning errors.

After "channel cleaning," the positioning error distribution of DAI UWB is generally optimized compared to the original channel, as some significant multipath reflection interferences that tend to produce noticeably abnormal large ranging error values have been cleaned up through channel estimation results. At this point, before channel cleaning, DAI UWB has only 55% of positioning errors less than 20 cm and 30% less than 10 cm. After channel cleaning, 80% of positioning errors were below 20 cm and 35% below 10 cm. Clearly, DAI UWB's channel estimation capability provides additional benefits for its positioning performance.

In the interest of rigor, this article also conducts a statistical analysis of the positioning error metrics, including the confidence interval (CI) analysis of the mean positioning errors achieved by various strategies, as well as the significance test of the statistical distribution characteristics of the positioning errors. After preliminary analysis using QQ plots, it can be determined that the positioning errors of each group conform

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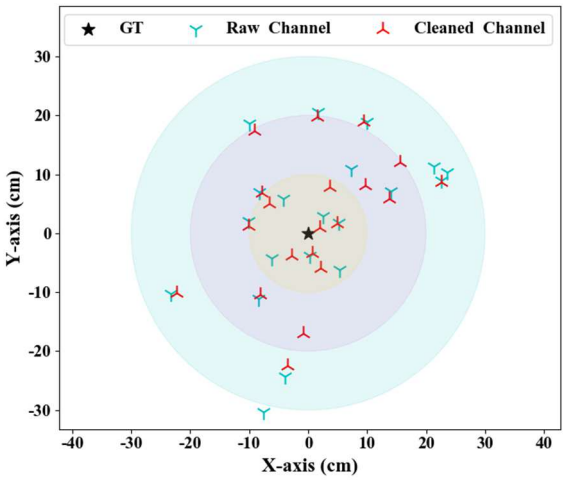


Fig. 11. Shooting target charts of positioning results, with three concentric circular target areas representing error bounds of 10, 20, and 30 cm, respectively.

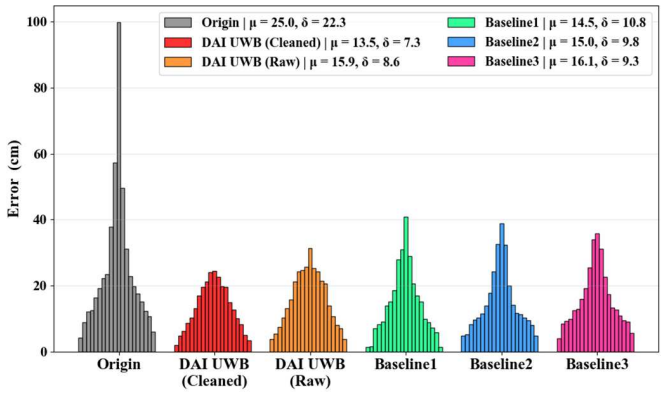


Fig. 12. Distribution characteristics of the positioning errors of different strategies with normal distribution characteristics.

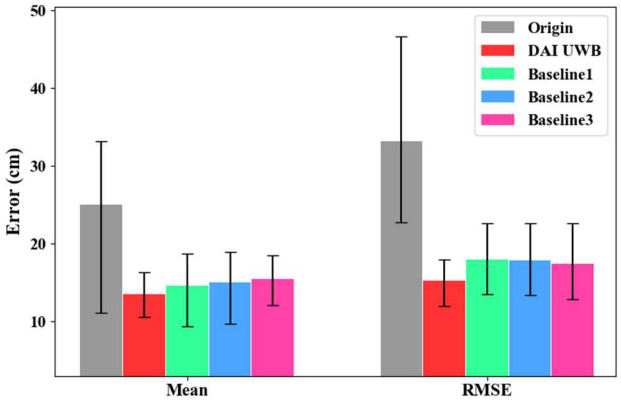


Fig. 13. Positioning performance indicators.

to certain normal distribution characteristics. Fig. 12 illustrates the distribution characteristics of the positioning errors of each group and the corresponding normal distribution parameters.

It can be intuitively observed that, after channel cleaning, the advantages of DAI UWB's channel estimation capability in mitigating positioning errors are more intuitively illustrated, and it exhibits the smallest mean value and overall distribution variance. Moreover, this article also conducted CIs analysis and significance tests on the positioning errors corresponding to each group of strategies. Fig. 13 presents a bar chart of the positioning error metrics, including error mean value and root mean square error (RMSE), with error bars (determined by the CIs). Table III records the 95% CIs of the positioning error metrics achieved by each strategy and the *p*-values of the T-tests, using the bootstrap method with 1000 times random sampling.

TABLE III
COMPARISON OF POSITIONING PERFORMANCE

Unit: cm	Mean & its 95% CIs	RMSE & its 95% CIs	P-value of T-test
Origin	25.0 (17.2, 35.6)	33.2 (19.7, 47.0)	1.0
DAI UWB (Cleaned)	13.4 (10.6, 16.7)	15.2 (12.4, 18.0)	1.0
Baseline1	14.5 (10.3, 18.9)	18.0 (13.0, 22.5)	1.0
Baseline2	15.0 (11.3, 19.1)	17.8 (12.8, 22.4)	1.0
Baseline3	16.1 (12.7, 20.3)	18.4 (13.9, 22.8)	1.0

It can be intuitively observed from Fig. 13 and Table III that the CI of the positioning error of DAI UWB is the most compact. This means that, from the perspective of statistical analysis, not only are the positioning error metrics (mean and RMSE) of the DAI UWB strategy the smallest but also the positioning availability of DAI UWB is the best among all strategies.

Moreover, after channel cleaning, the mean error and RMSE of DAI UWB are improved by approximately 10.7% and 15.6%, respectively. Compared with the original positioning error metric, DAI UWB has achieved improvements of 46.6% and 54.2% in mean error and RMSE, respectively. For a more in-depth discussion and analysis, please refer to Section V.

V. DISCUSSION

As shown in the evaluation results in Section IV, the policy network of DAI UWB for channel estimation demonstrates remarkable competitiveness on both the public eWINE dataset and our own. Moreover, it is distinguished by its lightweight characteristics, offering crucial practical value for edge devices. Both DAI UWB and all baselines demonstrate notable effects in compensating ranging errors in the original observations, leading to significant improvements in positioning accuracy. However, it is worth noting that while more accurate ranging values generally reduce positioning errors, the specific level of positioning error is not solely determined by ranging accuracy. It is also closely related to the spatial distribution of the anchors and the ill-condition issue of the positioning model due to the ranging error level.

As shown in Fig. 13, despite poorer ranging error compensation by Baseline1, it demonstrates performance that is comparable to the two baselines in terms of median error and RMSE metrics. This may be because, even if Baseline1

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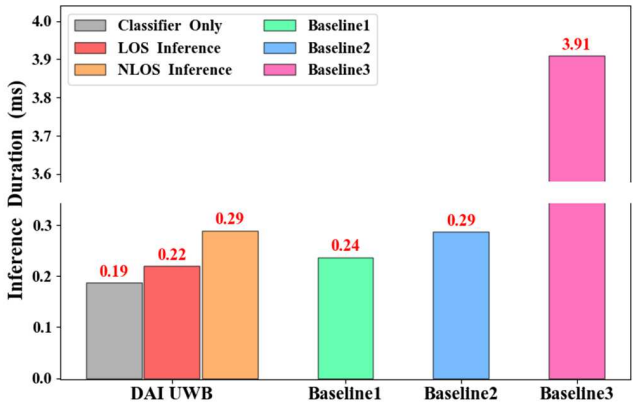


Fig. 14. Cost-time analysis at the edge.

does not achieve the optimal ranging value after compensation, it may also readily allow the NLS system described by (1) to approach the global optimum under the specific spatial distribution of anchors. This implies that, in many cases, a good anchor geometry distribution can provide a certain margin or tolerance for the fluctuations in ranging errors. This issue may be worth further in-depth exploration.

Moreover, DAI UWB provides users with channel estimation results for collected samples, offering additional benefits for the positioning system. For instance, in data contaminated by multipath effects, DAI UWB can identify some obvious abnormal channel samples and enhance positioning accuracy through data "cleaning." In many cases (although not in all situations), these cleaned data may be caused by significant multipath interference. This leads to their results that differ from the channel state of the same batch of data, thereby generating gross errors. Unlike end-to-end static models, which may underperform when facing multipath-contaminated or inaccurately labeled data, DAI UWB's dynamic nature allows it to adapt better to complex environments, yielding more reliable positioning results. Furthermore, DAI UWB outperforms [3] in channel estimation. Below is a discussion on this issue.

First, the model in [3], constrained by the limited computational resources of microprocessors and the on-chip deployment requirements of TinyML, did not fully utilize all RF channel features and the dual-branch structure employed in this article. Second, the dual-branch structure in this article has been optimized for targeted representation learning for high-level semantic features by separating RF channel and CIR features with distinct semantic meanings. This approach, which emphasizes the semantic distinction between different features, is more advanced than the simple concatenation of RF channel features and CIR features as a vector input [4].

Compared to recent studies, DAI UWB significantly reduces computational complexity while maintaining high performance. Many recent advances like [7], [8], [9], [10], achieve success in channel estimation and ranging error compensation, but at the cost of high computational complexity and hundreds of thousands, or even millions, of parameters. In contrast, DAI UWB employs DNN's idea to compensate for ranging errors differentially based on channel state evaluations from its policy network with 10 960 parameters, showcasing advantages in both efficiency and resource usage. For "simple" samples (deemed LOS channels), DAI UWB uses only RF channel features for ranging error compensation, with an inference parameter count of 12 217. For "complex" samples, both RF and CIR features are employed, utilizing 29 985 parameters. This "tailored" strategy enhances inference efficiency and reduces computational resource wastage. Moreover, although Baseline3 employs a CNN-LSTM network structure and achieves excellent performance in ranging error compensation, its model has the highest number of parameters among all the strategies discussed in this article, at 92 161. However, its largest parameter count does not translate into a significant performance advantage (see Fig. 7). While employing larger-scale deep learning models remains a promising approach to breaking through the precision bottleneck, for UWB, which is a micro-sized, low-cost commercial product, there exists a conflict between performance and practicality. The time-cost analysis shown in Fig. 14 can corroborate this viewpoint.

The cost-time analysis tests are conducted on a Raspberry Pi 4B, a commonly used edge device, with a memory capacity of 8 GB. For information regarding this edge device, please refer to its official website (https://www.raspberrypi.com/). The number of parameters for Baseline1 and Baseline2 are 13 769 and 28 295, respectively. Since there is no difference in the test results on the dataset between the Raspberry Pi 4B and the workstation, only the inference time metric is recorded here. We repeated the inference process ten times at the edge and recorded the time-cost metric as shown in Fig. 14.

Given that the Raspberry Pi 4B has a certain level of computational capability, running the DAI UWB model with lightweight nature for sample inference can be considered to have a very low latency. With a precision of two decimal places, for inference on a single sample, the average inference time for DAI UWB on each "simple" sample is 0.22 ms, and for each "complex" sample, it is 0.29 ms. Especially, if DAI UWB is used solely for channel estimation, the average inference time for each sample is 0.19 ms. The average inference time for a single sample of Baseline1 and Baseline2 is approximately 0.24 and 0.29 ms, respectively (with two significant digits of precision), which is significantly lower than the single-inference delay of Baseline3 by an order of magnitude (3.91 ms). It can be seen that if a deep learning model with much larger model parameters and complexity than Baseline3 is used, the inference delay on the edge side will correspondingly increase. In cases where there is a significant delay, it may even affect the usability of the UWB system in real-time measurement activities.

Finally, there is a discussion on the potential application prospects and further improvements of the DAI UWB work. We believe that DAI UWB technology should not be confined to channel estimation, ranging, and positioning tasks. Its broader potential applications should be explored. For example, applying DAI UWB in range-aided simultaneous localization and mapping (SLAM) [46] may lead to more reliable mapping capabilities. Additionally, given the robocentric visual-inertial odometry proposed in [47], DAI UWB could

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improve pose estimation and eliminate cumulative errors in such scenarios. Certainly, DAI UWB may also contribute to seamless indoor-outdoor positioning applications, combining with AI IMU [48].

Moreover, we believe that, the DAI UWB's design philosophy, based on DNN's idea, could inspire the development of other intelligent instruments. For instance, AI cameras could dynamically adjust computational resources to adapt to varying scenario complexities. Similarly, AI IMU technology could adopt the DNN approach to achieve more efficient dead reckoning. Additionally, given the role of attention mechanisms in enhancing deep learning models, the latest reported tensor product attention (TPA) [49] merits attention. This method dynamically factorizes tokens, obviating the need for the storage of complete static K and V vectors. By integrating adaptive attention weights, it achieves a 90% reduction in memory usage while maintaining strong performance, thus providing valuable insights for further refining DAI UWB.

VI. Conclusion

This article proposes DAI UWB, an integrated solution for UWB channel estimation and ranging enhancement based on dynamic AI approach. By leveraging DNN's philosophy, DAI UWB can automatically adjust the dynamic inference routing through channel estimation, enabling efficient ranging error compensation and contributing to improved positioning performance. Results demonstrate that DAI UWB excels in various complex indoor and outdoor settings, significantly reducing ranging errors and achieving high accuracy and robustness in positioning tasks. Moreover, DAI UWB's outstanding channel estimation capability provides an advantage in static positioning systems by filtering data contaminated by multipath effects, thereby offering reliable positioning support for practical applications.

The core advantage of DAI UWB lies in its capability to dynamically adjust the network structure and inference routing resources based on the complexity of the input data, thereby cutting down computational complexity while enhancing its inference efficiency. This design concept is not only applicable to UWB technology but can also be extended to other intelligent instruments that require high performance and efficiency. Thus, the design philosophy of DAI UWB offers valuable insights for the development of other AI-enabled intelligent instruments. In the future, we intend to further optimize the solution of DAI UWB and explore its potential applications in a wider range of scenarios to promote the continuous development of intelligent sensing technologies.

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Xu He (Graduate Student Member, IEEE) is currently pursuing the Ph.D. degree in instrumentation science and technology with the School of Instrument Science and Engineering, Southeast University, Nanjing, China.

He is with the School of Instrument Science and Engineering, Southeast University. He has authored six JCR Q1/Q1 TOP papers, and holds ten Chinese patents in the area of PNT and AI. His research interests include brain-inspired navigation, intelligent PNT, Internet of Things (IoT), and AGI.

Mr. He is a student member of CCF.



Xiaolin Meng was a Professor of Intelligent Mobility and a Theme Leader of positioning and navigation technologies with the University of Nottingham, Nottingham, U.K., until in May 2020. He is a Chair Professor of Intelligent Mobility with the School of Instrument Science and Engineering, Southeast University, Nanjing, China. He has authored more than 380 publications and undertaken a research portfolio of more than £ 20 m. His research interests mainly include intelligent transportation systems and smart cities, connected and autonomous vehicles, structural health monitoring of large infrastructure, and precision agriculture/livestock farming.

Dr. Meng is the first-ever Professor of Intelligent Mobility in U.K. He has been selected by U.K.'s Research Council as an Academic Fellow, and is a fellow of the RIN and IET, and a Chartered Engineer (CEng) of the U.K.'s Engineering Council. He is the Executive Editor, Associate Editor, Guest Editor, Editor, and reviewer for over 30 important international journals, including the RIN's "*Journal of Navigation*," Springer Nature's "*Satellite Navigation*" and "*Urban Lifeline*." His research achievements have been featured in 16 special reports by Science and Technology Daily and BBC. He has served as the Chairman for the Institute of Navigation (ION) session. He received the ION Best Paper Award and the 2019 U.K. Engineer "Collaborate to Innovate" Award. He is the Founder of the Sino-U.K. Geospatial Engineering Center and also the Founder and Chief Scientist of UbiPOS U.K. Ltd.



Lingfei Mo (Member, IEEE) is an Associate Professor with the School of Instrument Science and Engineering, Southeast University, Nanjing, China. He has authored over 100 technical publications in standard journals and conferences and has more than 40 Chinese patents in the area of Internet of Things (IoT) and AI. His research interests include IoT, AI, brain-inspired navigation, and brain-like computing.

Dr. Mo is an Associate Editor of IEEE JOURNAL OF RADIO FREQUENCY IDENTIFICATION (RFID).



Fanxing Yang is currently pursuing the M.S. degree in instrumentation science and technology with the School of Instrument Science and Engineering, Southeast University, Nanjing, China.

He is with the School of Instrument Science and Engineering, Southeast University. His research interests include Internet of Things (IoT), ultrawide band (UWB) positioning, and edge AI.



Youdong Zhang is currently pursuing the Ph.D. degree in instrumentation science and technology with the School of Instrument Science and Engineering, Southeast University, Nanjing, China.

He is with the School of Instrument Science and Technology, Southeast University. His research interests include brain-inspired navigation and brain-inspired computing.



Yue Wu is currently pursuing the M.S. degree with the School of Instrument Science and Engineering, Southeast University, Nanjing, China.

She is with the School of Instrument Science and Technology, Southeast University. Her research interests include brain-inspired navigation.

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